7.2.2 Symptoms

Signs and symptoms of this disease appear around 3-4 years of age. Affected infants may have hypoglycemia associated with seizures. High level of blood uric acid and hyperlipidemia are evident. Symptoms include low blood sugar, intolerance of fasting, frequent nose bleeds. Both liver cells and the cells of renal convoluted tubules are loaded with glycogen. Ketosis and hyperlipidemia are also present. Treatment includes intake of food high in starch

Figure.7.4 Von Gierke affected child

7.3 HEMOPHILIA

Clotting factors are a set of proteins that control bleeding. When a blood vessel is affected, the lining of the blood vessels contract to reduce the blood loss at the site of injury. Small blood cells called as platelets stick to the injured site and prevent blood loss. Inside the platelets, chemicals that attract other cells and make up a clump called as a platelet plug. At the site of injury, in the surface of platelets, clotting factors orchestrated by a set of proteins which form a coagulation cascade leading to the formation of fibrin clot. The fibrin clot acts as a mesh and stops bleeding.

Figure 7.5 Hemophilia

The term Hemophilia was first coined by Dr. John Conrad Otto, in the early 19th century. Hemophilia is an inherited bleeding disorder in which a person lacks or has low levels of certain proteins called "clotting factors" and the blood does not clot properly. There are nearly 13 factors involved in the mechanism of blood clotting. If any one or more of these factors are not synthesized adequately that results in the defect in blood clotting and thereby haemorrhage. A number of inherited deficiencies of the blood clotting factors are found in human and are collectively called as hemophilias. The clotting factors are presented in Table 7.1.

Clotting factor	Factor name	Source
I	Fibrinogen	Liver
II	prothrombin	Liver

XII U7 Inborn.indd 172 4/8/2020 3:22:38 PM